Homework 2
Datalink Layer

Release Date 1404/2/4
Deadline 1404/2/19

Theoretical Problems

This assignment is designed to provide you with deeper insights into the concepts of the Data Link Layer. If you score more than 50% of the total points, any additional points will be awarded as bonus.

Problem 1 (10 pts)

What is the maximum overhead in byte-stuffing algorithm? Explain your answer.

Problem 2 (20 pts)

A channel has a data rate of R bps and a propagation delay of ν km/s. The distance between the sending and receiving nodes is L kilometers. Nodes exchange fixed-size frames of B bits. Find a formula that gives the minimum 'sequence number' field size of the frame as a function of R, ν, B, and L (considering maximum utilization). Assume that ACK frames are negligible in size and the processing at the nodes is instantaneous.

Problem 3 (20 pts)

Frames of 1000 bits are sent over a 1-Mbps channel using a geostationary satellite whose propagation time from the earth is 270 msec. Acknowledgements are always piggybacked onto data frames. The headers are very short. Three-bit sequence numbers are used. What is the maximum achievable channel utilization for Stop-and-wait?

Problem 4 (20 pts)

The distance from earth to a distant planet is approximately 9×10^{10} m. What is the channel utilization if a stop-and-wait protocol is used for frame transmission on a 64 Mbps point-to-point link? Assume that the frame size is 32 KB and the speed of light is 3×10^8 m/s.

Problem 5 (30 pts)

In Figure 1, frames are generated at node A and sent to node C through node B. Determine the minimum data rate required between nodes B and C so that the buffers of node B are not flooded, based on the following:

- The data rate between A and B is 100 kbps.
- The propagation delay is for both lines.

- There are full duplex lines between the nodes.
- All data frames are 1000 bits long; ACK frames are separate frames of negligible length.
- Between A and B, a sliding-window protocol with a window size of 3 is used.
- Between B and C, stop-and-wait is used.
- There are no errors.



Figure 1: Sample Network with three nodes.

Submission

Please submit your answers to the problems in a `.pdf` file named **DN_HW2_studentID.pdf**, replacing `studentID` with your actual student ID.